

1 Conclusions

Chapter 1 framed this thesis as an end-to-end engineering pipeline (from a raw optical video archive to an uncertainty-aware design-triage tool) for marine multi-fuel pilot injection screening. The principal novelty of this thesis is not a new spray law or neural-network architecture, but an integrated, censoring-aware workflow that connects finite-FOV Mie observations, reduced-order trajectory targets, heteroscedastic surrogate learning, and geometric risk-oriented screening. This closing chapter synthesizes the core contributions, key scientific and empirical findings, evidence boundaries, and future validation priorities.

1.1 Novelty and Contributions

The primary contributions of this thesis are structured across three levels:

1. **Core Methodological Contribution:** An end-to-end censoring-aware learning chain that unifies finite-FOV observation auditing, Sigmoid-Gated Quarter-Root (SGQR) target construction, raw/teacher hybrid supervision, and heteroscedastic neural prediction. This formulation mitigates the risk of mistaking optical image window exit for physical spray saturation.
2. **Downstream Methodological Contribution:** A geometric screening interface that projects predicted penetration probability distributions into time-resolved wall-interception scores ($P_{\text{hit}}(t)$) and cumulative exposure E). This serves as a continuous geometric proxy for early-stage design triage ahead of computational fluid dynamics (CFD) simulation.
3. **Engineering Contribution:** An automated, CUDA-accelerated video processing and checkpointing pipeline capable of ingesting raw multi-hole Mie scattering archives (processing 2,992 recordings in approximately 72 minutes) into auditable plume-wise temporal observables.

1.2 Main Findings

First, finite-window censoring is a substantial property of the archive: 39.6% of the cleaned CDF penetration trajectories reach their nozzle-specific field-

of-view limit. The three-parameter SGQR fit addresses the resulting target-construction problem more robustly than the four-parameter two-branch candidate, increasing the post-fit-shielding pass rate from 61.3% to 90.0% while retaining a median accepted-fit RMSE of 1.27 mm. These results support its use as a smooth trajectory compressor and target generator; they do not validate physical continuation beyond the camera window.

Second, the time-resolved scaling analysis shows that the effective pressure-difference exponent is close to 0.5 over the well-supported early-to-mid transient window before moving towards the classical post-breakup asymptote. Relative to the $\Delta P^{0.25}$ alternative, the adopted $\Delta P^{0.5}$ scaling produces an approximately $6.6\times$ tighter collapse of the representative trajectory population. This finding provides the empirical basis for the amplitude scaling and residual-pressure feature design used by the surrogate.

Third, on the primary uncensored CDF protocol (540,367 held-out points per seed from the nine-nozzle archive), the production MLP achieves a five-seed mean RMSE of 4.185 mm and MAE of 2.978 mm. Under this dataset, protocol, and the refitted baseline definitions, this is a 58.9% RMSE reduction relative to Hiroyasu–Arai (10.192 mm) and a 54.6% reduction relative to Naber–Siebers (9.212 mm). On the same fixed table, the seed-42 SVGP reference attains lower RMSE and CRPS (4.087 mm and 2.063 mm, versus 4.185 mm and 2.301 mm for the MLP), whereas the MLP produces wider in-domain intervals ($1\sigma/2\sigma$ coverage 0.888/0.990 versus 0.717/0.946) and supports the onset output and regime-aware distillation chain used by the screening workflow. Because the seed protocols are unmatched, this comparison is descriptive rather than a statistical-significance claim, and conservative coverage is not a false-negative safety guarantee.

The five-fold LONO study further defines the transfer boundary. After excluding Nozzle 0, the four-fold mean RMSE is 7.00 mm for the MLP and 5.95 mm for the SVGP, indicating useful but imperfect transfer within the evaluated design lineage. On Nozzle 0, the corresponding errors rise to approximately 35 mm and 27 mm. Nozzle 0 is a factory-fresh reference injector outside the prototype design lineage, so this fold combines operating-condition extrapolation with an injector-population shift; it should not be interpreted as an ordinary in-family fold or generalized into a claim about all unseen hardware families.

Table 1 summarizes these primary findings alongside their quantitative evidence and operational scope.

Finding / Theme	Quantitative Evidence	Operational Scope / Boundary
FOV Right-Censoring SGQR Fit Robustness	39.6% of cleaned CDF trajectories reach the FOV limit Pass rate improved from 61.3% to 90.0% (median RMSE 1.27 mm)	Common to the finite-window Mie archive Trajectory compressor; wide-FOV continuation unvalidated
Transient Pressure Scaling Empirical Baseline Gain Model Trade-off	$\Delta P^{0.5}$ gives $\sim 6.6\times$ tighter collapse than $\Delta P^{0.25}$ MLP RMSE reduced by 58.9% (H-A) and 54.6% (N-S) SVGP RMSE/CRPS: 4.087 mm/2.063 mm; MLP: 4.185 mm/2.301 mm with wider coverage	Supported early-to-mid transient window Primary protocol and refitted baselines only Descriptive comparison with unmatched seed protocols
LONO Transfer Boundary	Excluding Nozzle 0: MLP 7.00 mm, SVGP 5.95 mm; Nozzle 0: MLP approx. 35 mm, SVGP approx. 27 mm	Nozzle 0 combines extrapolation and injector-population shift

Table 1: Summary of primary findings, quantitative evidence, and scope boundaries established in this thesis.

1.3 Evidence Boundary and Future Validation

All predictive claims remain limited to the room-temperature, non-reacting, constant-volume-vessel conditions and the Mie-defined liquid-envelope observables represented in the archive. The downstream projection additionally uses a fixed half-cone angle and simplified Gaussian spray geometry; it is not a phase-resolved engine impingement model. Within this operating scope, three evidence gaps require future experimental and computational validation:

- **Wide-FOV Extrapolation Validation:** The late-stage temporal extrapolation hypothesis of the SGQR target construction and knowledge distillation chain requires verification against wide-field single-plume optical recordings.
- **Physical Impingement Ground Truth:** The probabilistic wall-interception metric $P_{\text{hit}}(t)$ acts solely as a geometric proximity proxy; validating whether elevated contact scores correspond to actual

liquid droplet impingement or wall wetting requires matching high-temperature evaporative CFD or optical spray-wall interaction experiments.

- **Statistical Independence and Confidence Intervals:** Because frame-wise observations within a trajectory are temporally autocorrelated, future assessments should report confidence intervals clustered at the recording, operating-condition, and nozzle levels.

1.4 Closing Statement

By establishing an end-to-end, censoring-aware workflow, this thesis demonstrates how archival optical Mie scattering recordings can be systematically repurposed into an uncertainty-aware pre-screening tool. The resulting methodology provides an auditable, reusable framework with the potential to narrow candidate nozzle geometries and operating conditions prior to resource-intensive CFD simulations and engine validation trials.